

Aaryan Aggarwal

Web Developer • UI/UX Designer • React & FastAPI

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SUMMARY

Web developer and UI/UX designer pursuing a B.Tech in Computer Science, focused on building **responsive, motion-rich React interfaces** backed by clean **Python REST APIs** (FastAPI, Django REST). Designs in **Figma** before writing code, turning wireframes into accessible, pixel-accurate UIs. **Smart India Hackathon Grand Finalist** and two-time hackathon podium finisher, with shipped work in real-time video, multilingual AI and intelligent scheduling. Currently deepening FastAPI expertise (async endpoints, OAuth2/JWT auth, testing and deployment).

TECHNICAL SKILLS

Frontend	React.js, JavaScript (ES6+), HTML5, CSS3, Vite, React Router, Responsive & Mobile-First Design, Scroll/Motion Animation
Backend	FastAPI, RESTful API Design, JWT Authentication, RBAC, Pydantic, SQLAlchemy ORM
Design	Figma, UI/UX Design, Wireframing, Prototyping, Design Systems, Visual Hierarchy, Accessibility (WCAG basics)
Databases & Tools	PostgreSQL, Docker, Git, GitHub, Swagger/OpenAPI
Languages	Python, JavaScript, SQL, Java, C
AI & Data	RAG, NLP, Speech-to-Text/Translation APIs, NumPy, Pandas, OpenCV, Matplotlib

PROJECTS

Edu-Essence | Real-Time Multilingual Classroom Platform

[Code](#) • 2026

- React 18, Vite, JavaScript, React Router, Django REST, PostgreSQL, JWT, LiveKit (WebRTC), Whisper, Sarvam AI
- Built a classroom platform where **a teacher speaks once and every student hears the lesson live in their own language**, supporting **8 Indian languages** (Hindi, Marathi, Tamil, Telugu, Bengali, Kannada, Gujarati, English) in both online and in-person modes.
- Developed the **React + Vite frontend** with protected routes, JWT login/refresh and role-based teacher/student dashboards, integrating **LiveKit WebRTC video rooms** and a **20+ endpoint REST API**.
- Designed a **cinematic, scroll-driven landing page** in vanilla JS: Lenis smooth scrolling, a 3D “door-reveal” storytelling section and an interactive **3D teacher avatar**, with `prefers-reduced-motion` fallbacks for accessibility.
- Unified the online (video) and offline (in-person) classroom flows into one seamless experience, connecting the UI to a speech pipeline (Whisper STT → Sarvam AI translation → text-to-speech) with live transcripts.

SyncChron | AI-Powered Academic Scheduling Platform

2026

- FastAPI, PostgreSQL, React.js, Docker, NLP, RAG, Genetic Algorithm
- Architected an **AI-driven timetable generation engine** that resolves complex multi-variable academic constraints (faculty, rooms, slots) using **Genetic Algorithms**.
- Engineered **concurrent timetable generation supporting 50+ simultaneous requests**, ensuring high availability under real institutional workloads.
- Designed a scalable **FastAPI + PostgreSQL** backend, containerised with **Docker**, and integrated **RAG and NLP pipelines** for intelligent scheduling recommendations.
- Implemented **role-based access control (RBAC) with four user roles**, each served a tailored React dashboard.

ACHIEVEMENTS

- Grand Finalist, Smart India Hackathon (SIH)** — India’s largest national-level hackathon.
- 2nd Place, Build with Trae Hackathon** — AI-assisted product build challenge.
- 3rd Place, INNOVATEX’26** — built a **Smart Waste Management System**.